

Sri Lanka Institute of Information Technology



Lab Submission
03

IT24610814

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Mathematics for Computing | IT1130

B.Sc. (Hons) in Information Technology

Exercise 1:

1.

```
2<3
```

```
ans = 1
```

2.

```
>> 3>5
```

```
ans = 0
```

3.

```
>> 5>=sqrt(10^2/4)
```

```
ans = 1
```

4.

```
>> 3<=5
```

```
ans = 1
```

5.

```
>> a=3;
```

```
>> b=27^(1/3);
```

```
>> a<=b
```

```
ans = 1
```

6.

```
>> 2~=4
```

```
ans = 1
```

Exercise 2:

```
>> p=2>3;
```

```
>> q=4<5
```

1.

```
>> p||q
```

```
ans = 1
```

2.

```
>> p&&q
```

```
ans = 0
```

3.

```
>> ~p
```

```
ans = 1
```

4.

```
>> ~q
```

```
ans = 0
```

5.

```
>> (~p||q)&& q
```

```
ans = 1
```

6.

```
>> (p&&q)||(~q||p)
```

```
ans = 0
```

7.

```
>> (p||~q)&&(p||q)
```

```
ans = 0
```

Exercise 3:

a)

1.

```
>> hex2dec("D4")
```

```
ans = 212
```

2.

```
>> hex2dec("C41")
```

```
ans = 3137
```

3.

```
>> hex2dec("2F2C")
```

```
ans = 12076
```

b)

1.

```
>> dec2hex(12)
```

```
ans = C
```

2.

```
>> dec2hex(1234)
```

```
ans = 4D2
```

3.

```
>> dec2hex(534)
```

```
ans = 216
```

c)

1.

```
>> dec2bin(123)
```

```
ans = 1111011
```

2.

```
>> dec2bin(245)
```

```
ans = 11110101
```

3.

```
>> dec2bin(771)
```

```
ans = 1100000011
```

d)

1.

```
>> bin2dec("110010")
```

```
ans = 50
```

2.

```
>> bin2dec("1010")
```

```
ans = 10
```

3.

```
>> bin2dec("111000111")
```

```
ans = 455
```

e)

```
>> num=[123,54,223];
```

```
>> oct=dec2base(num,8)
```

```
oct =
```

```
173
```

```
066
```

```
337
```

```
>> hex=dec2base(num,16)
```

```
hex =
```

```
7B
```

```
36
```

```
DF
```

```
>> bin=dec2base(num,2)
```

```
bin =
```

```
01111011
```

```
00110110
```

```
11011111
```

Exercise 4:

```
temp = input('Enter temperature: ');  
rpre = input('Enter Reactor Pressure: ');  
water = input('Enter Cooling Water: ');
```

```
T=temp>115;  
P=rpre>15;  
W=water>120;
```

```
Condition1=T && ~W;  
Condition2=~T && P;  
Condition3=~W;
```

```
if Condition1 || Condition2 || Condition3  
    disp("Alarm ON");  
else  
    disp("Alarm Off");  
end
```